USER INTERFACE DESIGN

LABSHEET-3

1. Create a basic webpage with a div box that changes color at different screen widths using CSS media queries. (You *can use Breakpoints: 600px, 768px, 1024px)*

CODE:

```
<!doctype html>
<html>
  <head>
     <meta name = "viewport" content = "width=device-width, initial scale = 1.0"/>
   <style>
     div{background-color:blueviolet;color:white;}

     @media screen and (min-width:600px) and (max-width:768px){
              div{background-color:white;}
       }

    @media screen and (min-width:769px) and (max-width:1024px){
             div{background-color:powderblue;}
       }
   </style>
 </head>
<body>
  <div>
    <p>What is price of your blind eye ?</p>
    <h1>If you are justice, please do not lie .</h1>
  </div>
</body>
</html>
```

Output:

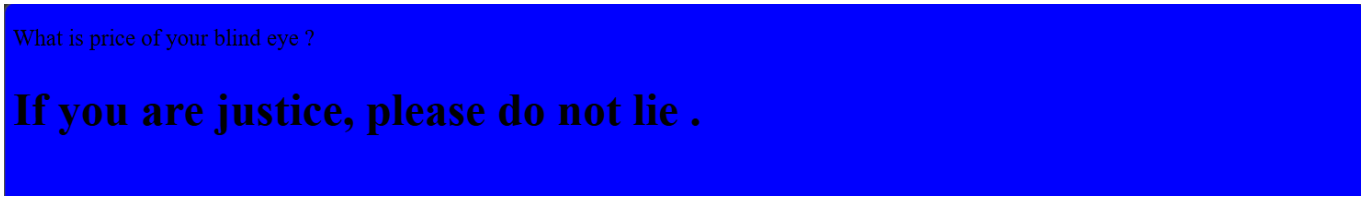
2. Create a web page with a paragraph of text.
Use **media queries** to change the font size and background color for the following screen widths:

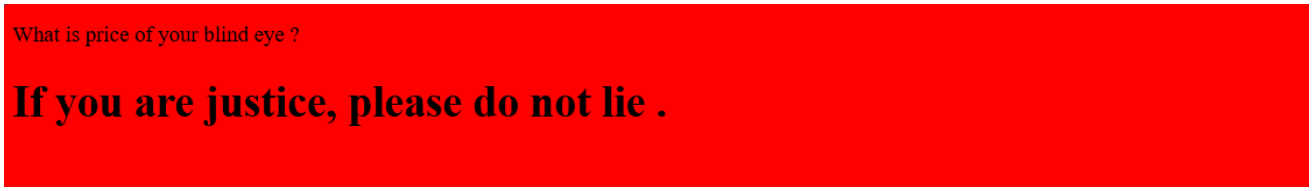
- Below 600px
- Between 601px and 1024px
- Above 1024px

CODE:

```
<!doctype html>
<html>
  <head>
     <meta name = "viewport" content = "width=device-width, initial scale = 1.0"/>
   <style>
        body{background-color:white;}

     @media screen and (max-width:600px){
        body{background-color:green; font-size:10px}

          }

     @media screen and (min-width:601px) and (max-width:1024px){
              body{background-color:red; font-size:15px}
       }

    @media screen and (min-width:1025px){
             body{background-color:blue; font-size:20px}
       }
   </style>
 </head>
<body>

    <p>What is price of your blind eye ?</p>
    <h1>If you are justice, please do not lie .</h1>

</body>
</html>
```

OUTPUT:





What is price of your blind eye ?

If you are justice, please do not lie .

3. Create a simple HTML page with a heading and paragraph. Add the following **viewport meta tag** in the <head> section:

<meta name="viewport" content="width=device-width, initialscale=1.0">
Test the page on a mobile screen (or use Chrome DevTools). **Observe and document** the difference in rendering **with and without** this meta tag.

CODE:

With meta-tag:

```
<!doctype html>
<html>
  <head>
     <meta name = "viewport" content = "width=device-width, initial scale = 1.0"/>
   <style>
        body{background-color:white;}

     @media only screen and (max-width:600px){
        body{background-color:purple;}

          }

     @media only screen and (min-width:601px){
              body{background-color:blue; font-size:15px}
       }
  </style>
 </head>
<body>


    <h1>If you are justice, please do not lie .</h1>
    <h3>What is the price for your blind eye ?</h3>
    <p>The answer is Justice.</p>

</body>
</html>
```

OUTPUT:

If you are justice, please do not lie .

What is the price for your blind eye ?

The answer is Justice.

If you are justice, please do not lie .

What is the price for your blind eye ?

The answer is Justice.

CODE:

Without meta-tag:

OUTPUT:

DIFFERENCE:

4. Testing with Images

Add a large image to your webpage (e.g., width: 100%). Without a viewport tag, see how the image renders.

Then add: <meta name="viewport" content="width=device-width, initial-scale=1.0">Note the difference in image behavior across devi screen600pxces.

CODE:

Without viewport:

```
<!doctype html>
<html>
  <head>

   <style>


  </style>
 </head>
<body>
   <h1>Alien Cyborg</h1>
  <img src = "https://i.redd.it/037ushhq5zze1.jpeg">

  </body>
</html>
```

With viewport:

```
<!DOCTYPE html>
<html>
<head>
<title>Image Test</title>

<meta = "viewport"  content = "width=device-width, initial scale = 1.0">

</head>

<body>

<h2>Large Image Test</h2>

<img src="https://i.redd.it/037ushhq5zze1.jpeg" style="width:100%;">

</body>
</html>
```

THE difference:

a. **In case of excluding meta tag :**

the image appears disproportionately large compared to actual screen width.

The webpage appears zoomed out.

b. **In case of including meta tag:**
The image appears in proportion to the screen width.
It's size appears to change as the screen width changes.

Note:

Please note that this difference can only be observed in a mobile phone.

In the desktop, it looks exactly the same.

5. Create a vertical navigation menu using an unordered list with at least 5 menu items.

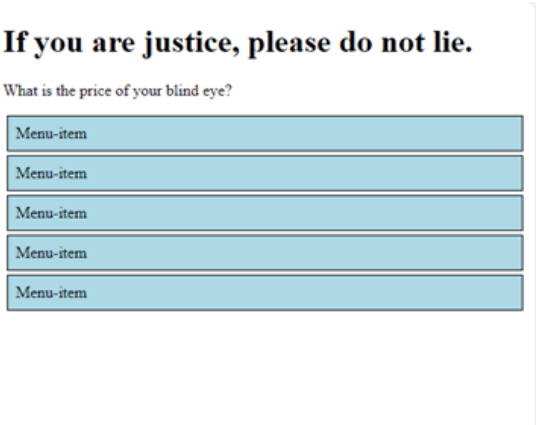
- By default, the menu should be displayed **vertically** with each item on a new line and a light blue background.
- When the screen width is **between 600px and 900px (using min-width and max-width media query)**, change the menu to display **horizontally**.
- Also change the background color to light green and align th
  e items in a single row.

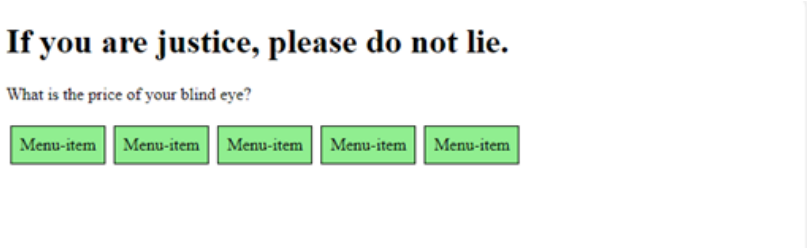
Use CSS media queries with both `min-width` and `max-width`.

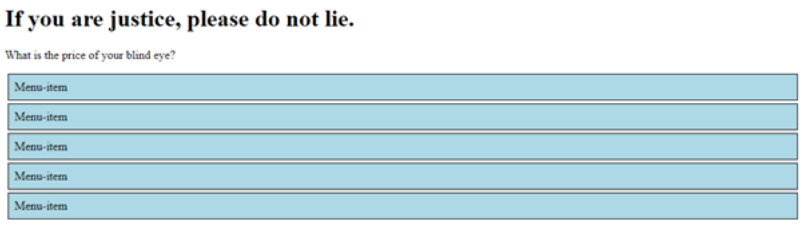
CODE:

```
<!doctype html>
<html>
<head>
 <title>LABSHEET 6</title>
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
 <style>
#list {margin:0; padding:0;}

.item {background-color:lightblue; border:1px solid black; list-style-type:none; margin:4px; padding:8px; display:block; }


@media screen and (min-width: 600px) and (max-width: 900px) {
 #list {display:flex; flex-direction:row;}
.item {background-color:lightgreen; margin:4px;}
}
</style>
</head>

<body>
<h1>If you are justice, please do not lie.</h1>
<p>What is the price of your blind eye?</p>

<ul id="list">
  <li class="item">Menu-item</li>
  <li class="item">Menu-item</li>
  <li class="item">Menu-item</li>
  <li class="item">Menu-item</li>
  <li class="item">Menu-item</li>
</ul>

</body>
</html>
```

OUTPUT:







6. Width with % vs px

Create a container with a background color and some text inside it.
Try setting its width to:

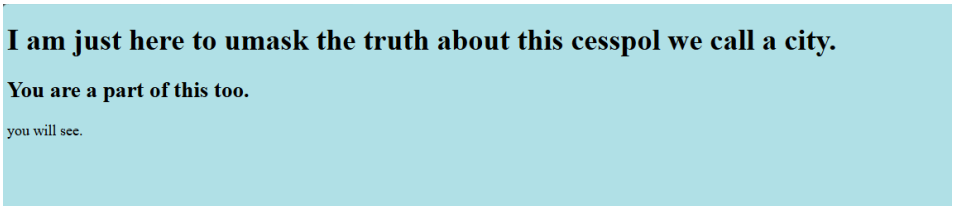
- 600px
- 80%

Resize the browser window and **observe**

how the layout behaves. Document the difference.

OUTPUT:

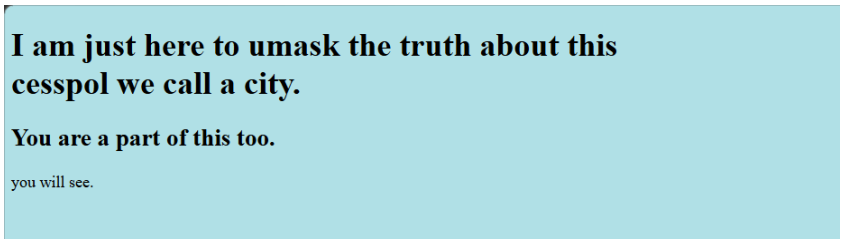
Default :



CODE(with px):

```
<!doctype html>
<html>

 <head>
  <title>LABSHEET-3</title>
    <style>
      div{width:600px;}
    </style>
  </head>

   <body style = "background-color:powderblue;">
<div>
   <h1>I am just here to umask the truth about this cesspol we call a city.</h1>
   <h2>You are a part of this too.</h2>
   <p>you will see.</p>
</div>
</body>
</html>
```

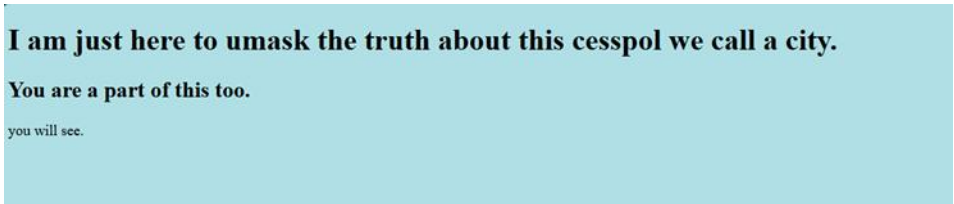
OUTPUT:



CODE(with %):

```
<!doctype html>
<html>

 <head>
  <title>LABSHEET-3</title>
    <style>
      div{width:80%;}
    </style>
  </head>

   <body style = "background-color:powderblue;">
<div>
   <h1>I am just here to umask the truth about this cesspol we call a city.</h1>
   <h2>You are a part of this too.</h2>
   <p>you will see.</p>
</div>
</body>
</html>
```

OUTPUT:



The difference:

In case of width, the lengthy sentence appears to have break somewhere. When resizing the browser window, we don’t see the sentence adjusting in proportion to it.

In case of %, the lengthy sentence does not break. When resizing the browser window, we see the sentence adjusting in proportion to it.

7. Font Sizes with em vs rem vs px

Create a paragraph and heading inside a parent container.
Try setting font sizes using:

- px (e.g., 18px
- em (e.g., 1.2em)

- rem (e.g., 1.5rem)

Change the root (html) font size and observe how **em and rem behave differently**

CODE:

With 18 px:

```
<!doctype html>
<html>

 <head>
  <title>LABSHEET-3</title>
    <style>
      div{font-size:18px;}
    </style>
  </head>

   <body style = "background-color:powderblue;">
<div>
   <h1>I am just here to umask the truth about this cesspol we call a city.</h1>
   <h2>You are a part of this too.</h2>
   <p>you will see.</p>
</div>
</body>
</html>
```

OUTPUT:

I am just here to umask the truth about this cesspol we call a city.

You are a part of this too.

you will see.

With 1.2 em:

```
<!doctype html>
<html>

 <head>
  <title>LABSHEET-3</title>
    <style>
      div{font-size:1.2em;}
    </style>
  </head>

   <body style = "background-color:powderblue;">
<div>
   <h1>I am just here to umask the truth about this cesspol we call a city.</h1>
   <h2>You are a part of this too.</h2>
   <p>you will see.</p>
</div>
</body>
</html>
```

OUTPUT:

I am just here to umask the truth about this cesspol we call a city.

You are a part of this too.

you will see.

With 1.5rem:

```
<!doctype html>
<html>

 <head>
  <title>LABSHEET-3</title>
    <style>
      div{font-size:1.5rem;}
    </style>
  </head>

   <body style = "background-color:powderblue;">
<div>
   <h1>I am just here to umask the truth about this cesspol we call a city.</h1>
   <h2>You are a part of this too.</h2>
   <p>you will see.</p>
</div>
</body>
</html>
```

OUTPUT:

I am just here to umask the truth about this cesspol we call a city.

You are a part of this too.

you will see.

The difference:

px (pixels)

Fixed size

Doesn’t change

em

Depends on the parent element

Changes if the parent size changes

rem

Depends on the root (html)

Changes if the root size changes

8. Using vh and vw for Fullscreen Sections

Create a <section> that fills:

- **100vh** (viewport height)
- **100vw** (viewport width)

Add a background image or color and center some text inside it.
Observe how it always fills the screen height/width regardless of content.

CODE:

```
<!doctype html>
<html>

 <head>
  <title>LABSHEET-3</title>
    <style>
      section{width:100vw; height:100vh;}
    </style>
  </head>

   <body style = "background-color:powderblue;">
<section>
   <h1>I am just here to umask the truth about this cesspol we call a city.</h1>
   <h2>You are a part of this too.</h2>
   <p>you will see.</p>
</section>
</body>
</html>
```

OUTPUT:

If you put 100, the section will fill the whole screen.

If you put a smaller value like 20, the section doesn't fill the whole screen and appears to break.

CSS Flexbox & Grid (Home work)

Create a responsive webpage layout using **CSS Grid and Flexbox**.

- Use **CSS Grid** to design the overall page layout with the following sections:
  Header, Navigation, Main Content, Sidebar, and Footer.
- Arrange the layout in two columns (Main Content and Sidebar side-by-side) for screens above 768px.
- For screen width below 768px, change the layout to a single-column structure.

Inside the **Navigation section**, use **Flexbox** to:

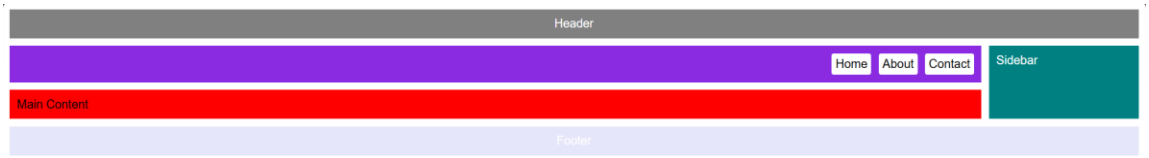
- Display menu items horizontally.
- Align them to the right.
- For screen width below 600px, change the navigation items to a vertical layout.

Use appropriate spacing and background colors for clarity.

OUTPUT:

```html
<!DOCTYPE html>
<html>
<head>
  <style>
    body {
      margin: 0;
      font-family: Arial, sans-serif;
    }

    .container {
      display: grid;
      grid-template-columns: 1fr 200px;
      grid-template-areas:
        "header header"
        "nav sidebar"
        "main sidebar"
        "footer footer";
      gap: 10px;
      padding: 10px;
    }

    header {
      grid-area: header;
      background: gray;
      color: white;
      padding: 10px;
      text-align: center;
    }

    nav {
      grid-area: nav;
      background: blueviolet;
      display: flex;
      justify-content: flex-end;
      gap: 10px;
      padding: 10px;
    }

    nav a {
      color: black;
      text-decoration: none;
      padding: 5px;
      background: white;
      border-radius: 3px;
    }

    main {
      grid-area: main;
      background: red;
      padding: 10px;
    }

    aside {
      grid-area: sidebar;
      background: teal;
      color: white;
      padding: 10px;
    }

    footer {
      grid-area: footer;
      background:lavender;
      color: white;
      text-align: center;
      padding: 10px;
    }

    @media (max-width: 767px) {
      .container {
        grid-template-columns: 1fr;
        grid-template-areas:
          "header"
          "nav"
          "main"
          "sidebar"
          "footer";
      }
      nav {
        justify-content: flex-start;
      }
    }

    @media (max-width: 599px) {
      nav {
        flex-direction: column;
        align-items: flex-start;
      }
    }
  </style>
</head>
<body>

  <div class="container">
    <header>Header</header>

    <nav>
      <a href="#">Home</a>
      <a href="#">About</a>
      <a href="#">Contact</a>
    </nav>

    <main>Main Content</main>

    <aside>Sidebar</aside>

    <footer>Footer</footer>
  </div>

</body>
</html>
```

OUTPUT:


Header
Home
About
Contact
Sidebar
Main Content
Footer

Header
Home
About
Contact
Main Content
Sidebar
Footer

Header
Home
About
Contact
Main Content
Sidebar
Footer